International Baccalaureate
Extended Essay

Topic: Physics

Investigation of the effect of surface tension on the vibration of musical drums

Research question:

What are the effects of surface tension on the vibration mode and frequency of drum heads?

Word count: 3879

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1. Introduction

The principle of acoustic instrument is a complex topic; thus, even though oscillation is learned in IB physics, it hardly covers its application on musical instruments. However, in real-life, various acoustic instruments rely on oscillation to phonate, including plucked string instruments, bowed string instruments, wind instruments, and percussion instruments, yet from oscillation of different elements. Therefore, as an amateur drummer, I thought it would be interesting to explore through the acoustic vibration of drums. Since the current research progress on drum vibration modes reaches the relationship between drum length and vibration modes, but seldom investigates the effect of surface tension, so I decided to dig deeper into this section.

This essay is an effort to solve the research question "What are the effects of surface tension on the vibration mode and frequency of drum?". The main objective of this essay is to first theoretically verify that the vibration mode and frequency of a single drum head will alter with a varying surface tension, then use a Finite Element Analysis (FEA) software *COMSOL Multiphysics* to simulate the vibration of drum head after excitation under several surface tensions, and finally experimentally investigate the effect of an altering surface tension on the vibration of drum heads. The experiment will use the Electronic Speckle-Pattern Interferometry system that optically records the simultaneous vibration modes, because the subtle oscillation mode and amplitude can hardly be captured by camera.

2. Investigation of the relevance between surface tension and vibration mode of single plate after excitation

2.1 Vibration mode

The vibrational modes of a circular membrane comprise of two elements and is denoted by *(m, n)*, where *m* is the number of nodal diameters and *n* is the number of nodal circles. Take drum as an example. Since drum already has its drum head clamped onto the wooden body, this bearing edge will count as one circular node, so that drum inherently have at least one circular node.

Figure 1 below is the illustration of several vibration modes on circular plate, where the circles indicate the circular nodes and the straight lines represent the nodal diameters. Also, during oscillation, each mode will produce a unique range of frequencies, as illustrated by the number blow each illustration--it measures how many times larger the frequency of this mode is, comparing to that of the (0,1) mode.

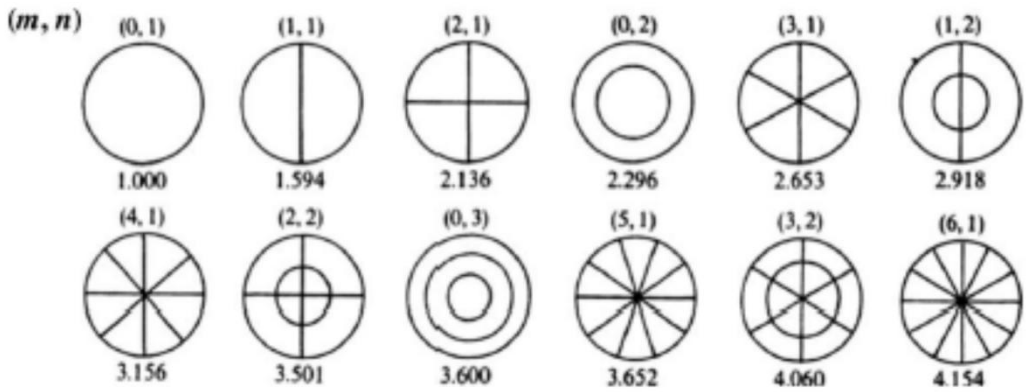
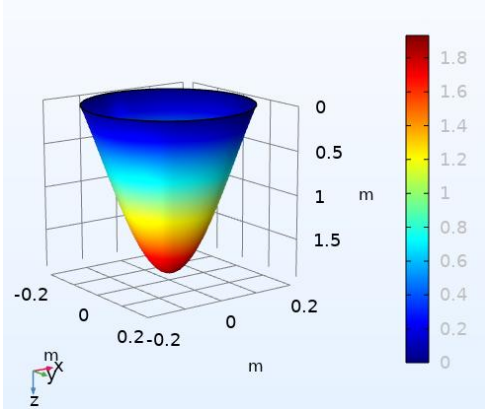
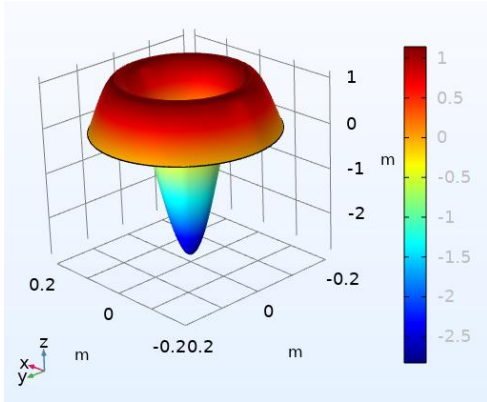


Figure 1. The illustration of 12 different vibration modes of circular membrane

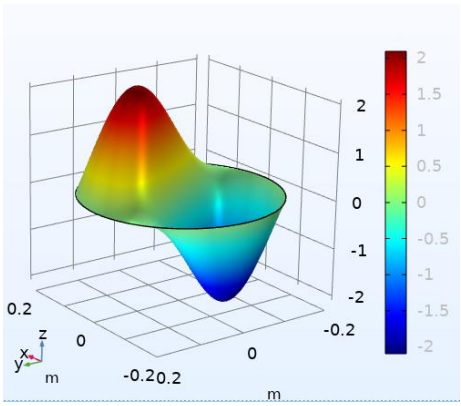
By using FEA software *COMSOL Multiphysics* to model the natural oscillation of a piece of drum head, I solved the first four types of vibration modes.



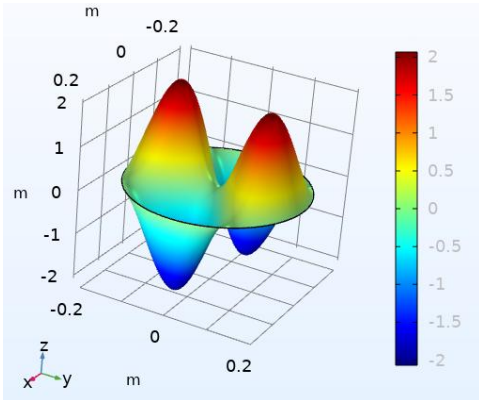
(a) Mode (0, 1)



(b) Mode (0, 2)



(c) Mode (1, 1)



(d) Mode (2, 1)

Figure 2. Image of the vibration pattern of modes (0, 1), (0, 2), (1,1), (2,1).

In fact, different vibration modes will account for different forms of sound: the modes consisting of no nodal diameters but only circular nodes, i.e. (0,*n*), generally contribute to the "beat" sound from drum, while the modes constituted of nodal diameter basically account for the acoustic "pitch".

2.2 Theoretical evidence of relationship between surface tension and vibration mode

The 2D wave function for transverse wave on circular membrane is a function that predicts the location of every particle on the membrane at each time interval. The general concept is that particles in the radial (r) direction follow the pattern of Bessel function, while cosine-type function accounts for the azimuthal (φ) direction:

$$\Psi_{m,n}^{displacement}(r,\varphi,t) = A_{m,n} J_m(k_{m,n} r)\cos(m\varphi)\cos(\omega_{m,n} t) \qquad [2.1]$$

Where r represents the radial displacement, φ represents the azimuthal displacement, t represents the time interval, $A_{m,n}$ is the vibration amplitude, J_m is the Bessel function, $k_{m,n}$ is the wavenumber, and $\omega_{m,n}$ is the angular velocity of the membrane vibration. Also, m and n are determined by the number of nodes and circular nodes respectively, that will be illustrated in vibration mode later.

From the wave function in Equation [2.1], we can catch a glimpse on the effect of surface tension on the particle displacement on the drum head: since the surface tension will influence the angular velocity of the wave propagation on the membrane, it may alter the "$cos(\omega_{m,n}\, t)$" part, thus influencing the overall particle displacement.

Digging deeper into Equation [2.1], we can divide it into products: overall vibration amplitude $A_{m,n}$, Bessel function $J_m(k_{m,n} r)$, cosine function $\cos(m\varphi)$, and another cosine function $\cos(\omega_{m,n} t)$.

2.2.1 Amplitude $A_{m,n}$

The amplitude of vibration is inversely correlated to the tension exerted on the drum head, because the tighter the drum head, the stronger force required to cause greater drum head displacement.

However, we can ignore the amplitude, as it has no connection with vibration modes, but only accounts for the proportional change of vertical displacement.

2.2.2 Bessel function $J_m(k_{m,n}r)$

Bessel function of the first kind is defined as the solutions to the Bessel differential equation [2.2], and explains phenomena including wave propagation.

$$x^2 \frac{d^2 y}{dx^2} + x\frac{dy}{dx} + (x^2 - m^2)y = 0 \qquad [2.2]$$

The graph of Bessel function also alters with the number of nodal diameter during vibration *m*, as shown in Figure 3 below:

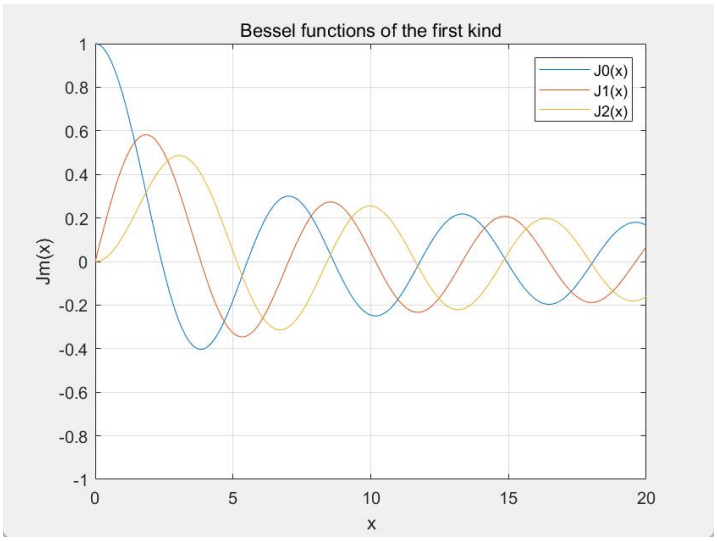


Figure 3. Bessel function of the first kind with m=1,2,3. Drawn by Matlab, with codes attached to Appendix 1

The Bessel function $J_m(k_{m,n}r)$ contains the independent variable *r*, meaning that the function predicts the particle movement in radial directions.

As $k_{m,n}$ represents the wavenumber, it can be denoted by:

$$k_{m,n} = \frac{2\pi}{\lambda_{wavelength}} = \frac{2\pi}{\frac{DrumDiameter}{NumberOfPeriod}} \qquad [2.3]$$

Therefore, the displacement of particles in radial directions only bears the influence of vibration modes *(m, n)* so far.

2.2.3 Cosine function (*φ* related) $\cos(m\varphi)$

Since this cosine function includes an independent variable *φ*, it means that the particle movement in azimuthal directions is predicted by a cosine function.

2.2.4 Cosine function (tension related) $\cos(\omega_{m,n}t)$

In order to find the formular relationship between angular velocity $\omega_{m,n}$ and surface tension, we first expand the ω into

$$\omega = 2\pi f = \frac{2\pi}{\lambda} \cdot \lambda f \qquad [2.4]$$

substituting $\frac{2\pi}{\lambda}$ with $k_{m,n}$, from Equation [2.3], and λf with wave propagation speed, to get:

$$\omega = k_{m,n} v_{WaveSpeed} \qquad [2.5]$$

Then, as the speed of transverse wave propagation is $v_{WaveSpeed} = \sqrt{\frac{T_1}{\sigma}}$, where T represents the surface tension per unit length (N m^{-1}) and σ represents the areal mass density of the membrane (kgm^{-2}), substituting this gets:

$$\omega = k_{m,n}\sqrt{\frac{T_1}{\sigma}} \qquad [2.6]$$

Therefore, the original cosine function $\cos(\omega_{m,n}t)$ finally transforms into a tension related formula $\cos(k_{m,n}\sqrt{\frac{T_1}{\sigma}} \cdot t)$.

As a functional coefficient in the wave function, the value of this cosine function will impose impact on the results of other components including the *r* related Bessel function and *φ* related cosine function, which relate to particle movements in both radial and azimuthal directions.

2.3 *COMSOL* simulation -- motivated vibration mode under different surface tension

From 2.2, the influence is theoretically proved. In order to verify this, I have modeled a two-dimensional drum head with three different magnitudes of surface tension (per unit length), varying from 11500 N m^{-1}, 23000 N m^{-1}, to 46000 N m^{-1} in Finite element analysis (FEA) software *COMSOL Multiphysics*.

Then, I monitored a drumstick with 4mm-square striking area, and used a force of 6.24×10^4 m s^{-2}, i.e. a force of 1N, to strike onto the center of each drum head.

Their vibration patterns after 0.02 seconds, which is the second frame after excitation, are shown in Figure 4:

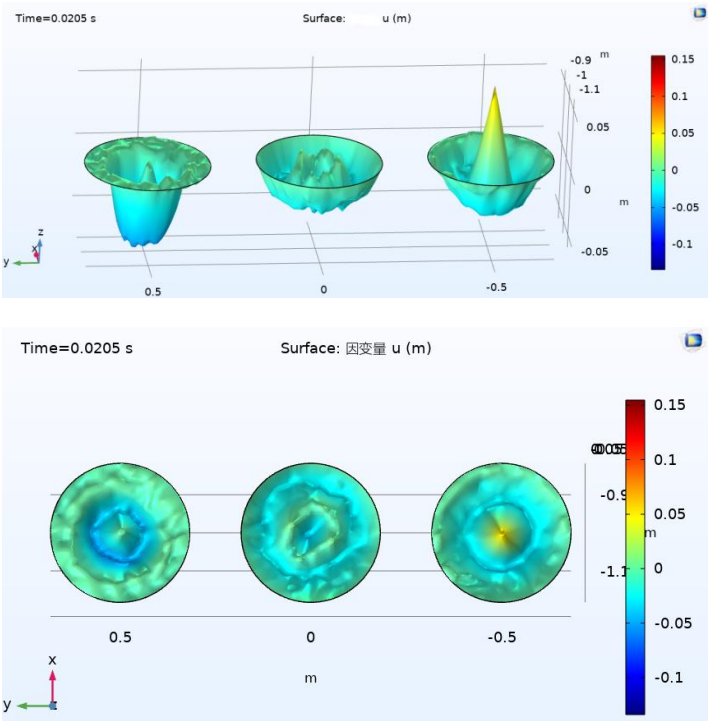


Figure 4. The vibration pattern after excitation of 6.24E04 ms^{-2} on every drum head, with surface tension (per unit length) of 11500 N m^{-1} (left), 23000 N m^{-1} (middle), and 46000 N m^{-1} (right) respectively.

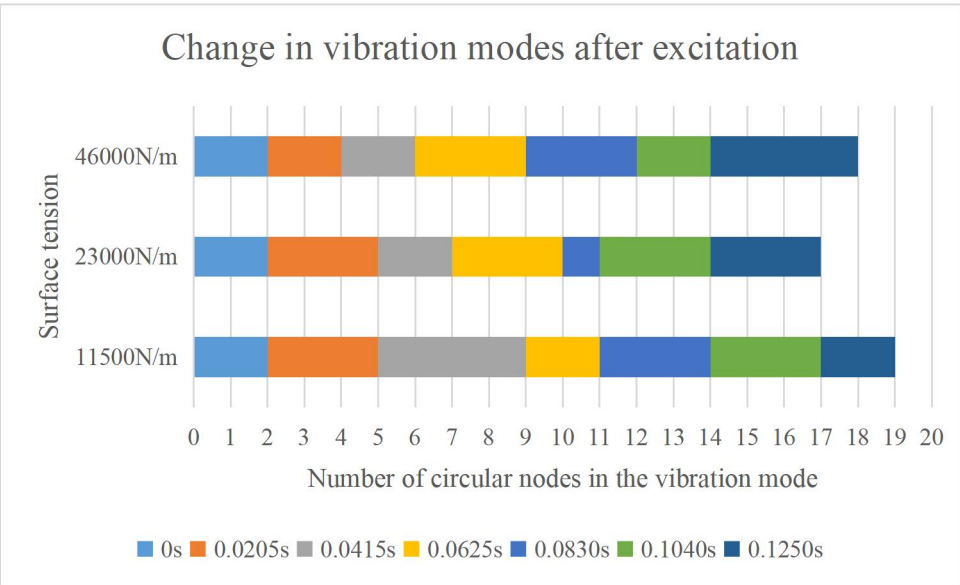
From Figure 4, we can generally distinguish the vibration modes of each drum head: the left one with 11500 N m^{-1} surface tension vibrates in (0,3) (since this left one has the smallest tension, i.e. smallest frequency, its oscillation is not complete in this frame, but it will develop into (0,4) in the next frame); the middle one with 23000 N m^{-1} surface tension vibrates in (0,3); the left one with 46000 N m^{-1} surface tension vibrates in (0,2).

However, the drum heads will not maintain the same mode of vibration forever. Instead, the pattern will alter roughly every one to two oscillation period. For example, in Table 1 are part of the data retrieved from the other frames of the simulation:

Table 1. The vibration modes of the three plates under sequential frames from *COMSOL* simulation

Frame Number / Surface Tension	0s	0.0205s	0.0415s	0.0625s	0.0830s	0.1040s	0.1250s
11500N/m	(0,2)	(0,3)	(0,4)	(0,2)	(0,3)	(0,3)	(1,2)
23000N/m	(0,2)	(0,3)	(0,2)	(0,3)	(0,1)	(0,3)	(0,3)
46000N/m	(0,2)	(0,2)	(0,2)	(0,3)	(0,3)	(0,2)	(0,4)

Figure 5. The change in the number of circular nodes in the vibration modes under different tensions (Data from Table 1)



3. Investigation of the relevance between surface tension and vibration frequency of single plate after excitation

3.1 Theoretical evidence of vibration eigenfrequency under different surface tension

3.2 *COMSOL* simulation -- vibration eigenfrequency under different surface tension

From 3.1, the theoretical mutual relationships between vibration mode, eigenfrequency, and surface tension are derived. In order to verify this, I have again modeled three two-dimensional drum heads with three different magnitudes of surface tension (per unit length), varying from 11500 N m^{-1}, 23000 N m^{-1}, to 46000 N m^{-1} in Finite element analysis (FEA) software *COMSOL Multiphysics*. The drum head has a radius of 0.1825m, and the material property applies a density (rho) of 1350 kgm^{-3}, a Young's modulus (E) of 9 GPa, and a Poisson's ratio (ν) of -0.7.

By adding Eigenfrequency Study in Solid mechanics Physics module, I am able to witness the vibration mode of each plate under non-trival frequencies. In the course of trying different non-trival frequencies, I recorded all frequencies that produce meaningful vibration mode patterns, such as the (3,2) mode illustrated below. From Figure 6, we can witness the similar vibration patterns in all three plates, except their eigenfrequency varies with their surface tension.

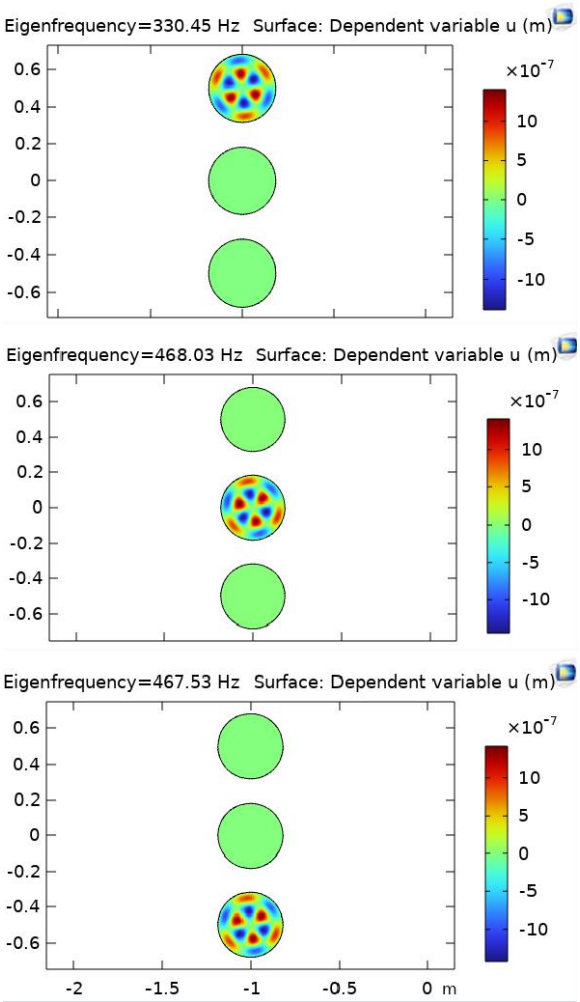


Figure 6. *COMSOL* simulated membranes in surface tensions of 11500 N m^{-1}(top), 23000 N m^{-1}(middle), and 46000 N m^{-1}(bottom) vibrating in eigenfrequencies written on the left. The horizontal and vertical axis denotes negligible positions of the membranes

Along with other collected simulation results (in Appendix 2), I summarized a table involving the eigenfrequencies of different vibration modes for the three plates.

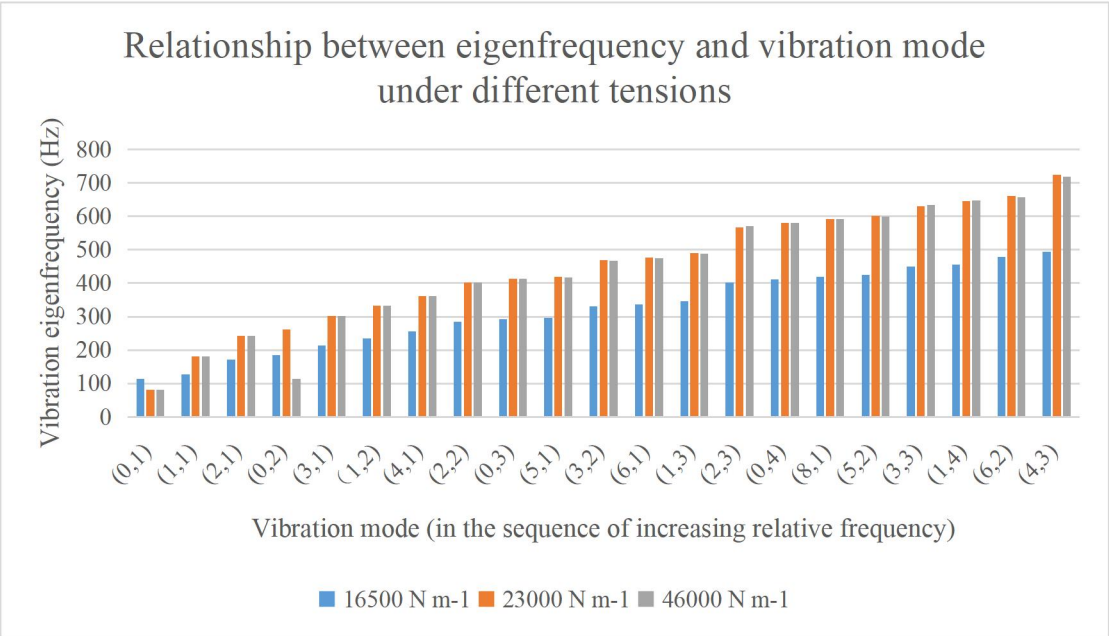
Table 2. Vibration modes with their eigenfrequencies under three surface tensions in *COMSOL* simulation

Vibration Mode / Eigenfrequency [Hz] / Tension [N m^{-1}]	**(0,1)**	**(0,2)**	**(0,3)**	**(0,4)**	**(1,1)**	**(1,2)**	**(1,3)**	**(1,4)**
11500	113.85	185.12	292.70	410.66	128.31	235.75	345.33	456.45
23000	80.505	261.85	413.72	579.65	181.47	333.52	489.09	646.29
46000	80.505	113.85	413.91	580.30	181.46	333.41	487.7	646.5
	(2,1)	**(2,2)**	**(2,3)**	**(3,1)**	**(3,2)**	**(3,3)**	**(4,1)**	**(4,3)**

11500	172.10	284.22	402.57	214.05	330.45	448.80	255.03	494.33
23000	243.41	401.64	566.84	302.82	468.03	630.95	361.10	723.31
46000	243.38	401.83	570.35	302.70	467.53	633.09	360.63	717.65
	(5,1)	**(5,2)**	**(6,1)**	**(6,2)**	**(8,1)**			
11500	295.73	424.42	336.17	477.92	418.40			
23000	418.33	600.86	476.00	660.45	591.81			
46000	418.09	599.18	475.38	657.89	591.68			

In Figure 7 below, I exhibits the data in Table 2 in a bar chart, so that the variations are clearer to see.

Figure 7. Bar graph representing the relationship between vibration eigenfrequency and vibration modes under different tensions (Data from Table 2)



When it comes to the horizontal comparison, we set tension as constant, and compare between vibration mode and frequency. First of all, the *n* value for every group of same value of *m*, shows a positive correlation between the eigenfrequency, which is consistent with the theoretical trend. Second, as the vibration modes in the graph take

the order of size of the relative frequency in Figure 1, we can see from the rising trend of all eigenfrequencies that at every constant tension, the vibration frequency is positively correlated with each vibration mode's relative frequency, except for one outlier (0,2) of 46000 N m^{-1}.

However, when vertically comparing the frequency of same vibration modes in plates of different tension, we can witness a clear inconsistency with the theoretical trend. According to the theory, $f_{m,n} \propto \sqrt{T_l}$, but most frequencies of group 23000 N m^{-1} and 46000 N m^{-1} exhibit an abnormal similarity, which disobey the theoretical expectation.

Considering the fact that when vertically comparing the 115000 N m-1 group with the other two higher-tension groups, the relationship generally follows the theory except for the (0,1) mode, we can speculate that the theory does not apply to the plate with overwhelming surface tension, causing the resultant frequency of the last two groups to be similar.

3.3 *COMSOL* simulation -- vibration frequency under different surface tension in stimulated vibration

The vibration frequency can be practically examined by horizontally comparing how fast the vibration modes are changing in the simulated plate vibration. I still used the

same parameters for the drum head membrane tension and the striking section as illustrated in Section 3.3.

To clarify my criterion for distinguishing a vibration period, I will consider the end of a period to be when the particles on the drum head return to the position at the start of the period. For instance, I have captured the photo of a series of drum head vibration after struck, which is shown in Figure 8 below.

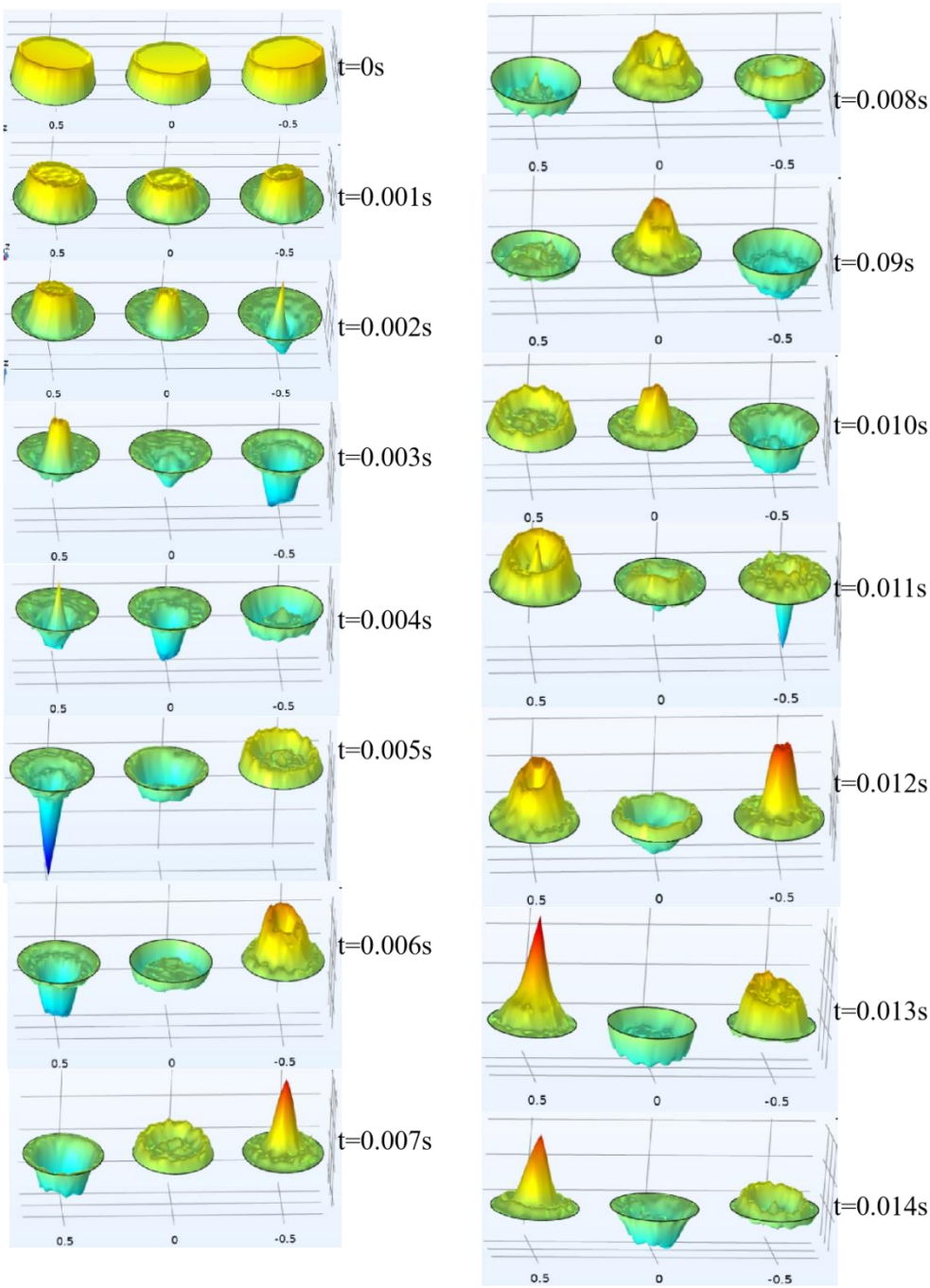


Figure 8. Shots of drum head over 0.014s after an excitation of 6.24E04 ms^{-2} on every drum head, with surface tension (per unit length) of 11500 N m^{-1} (left column), 23000 N m^{-1} (middle column), and 46000 N m^{-1} (right column) respectively.

From frame t=0s and t=1s, we know that the vibration starts from the highest positive amplitude and gradually decreases, which seems like a “cosine function”. Therefore, the point-in-time for this period to end is when the highest positive amplitude is reached again.

At t=0.007s, we can observe the highest positive amplitude appearing again in the membrane of 46000 N m^{-1} surface tension (right column), as the amplitude drops at t=0.008s again. Therefore, the period for the first vibration for 46000 N m^{-1} tension is 0.007s.

Similarly, we can distinguish that the period for the 23000 N m^{-1} tension drum head (middle column) is 0.009s, and that for the 11500 N m^{-1} tension drum head is 0.013s (left column).

Therefore, with the multiple growth of the surface tension, their period of the first vibration is also increasing.

4. Experimentally verifying the vibration modes under different surface tensions using interferometry

Since the vibration pattern of the drum head is too subtle to be observed by bare eye or camera, I relied on the principle of interference to magnify the change in drum head.

An interferometer is a device that measures changes in optical or electromagnetic waves by superimposing two or more waves to detect changes in their relative phase. This allows it to detect subtle changes in the properties of the wave, such as its wavelength, frequency, or amplitude. For example, an optical interferometer can be used to detect small changes in the shape or refractive index of a material by illuminating it with a laser beam and comparing the phase shift of the light that passes through it to a reference beam, allowing it to make highly sensitive observations

4.1 Experiment design

The first interferometer I built is a Michelson interferometer. A basic Michelson interferometer consists of two mirrors, a light source, and a beam splitter. Light from the source is split into two separate beams by the beam splitter and each beam is directed towards a mirror. The beams are then reflected back towards the beam splitter and recombined. The interference pattern that is produced can be used to measure the difference in optical path length between the two beams.

The principle is, when we keep one mirror stationary and move the other, the change in distance between the mirror and the beam splitter will induce phase change in that beam of laser, thus changing the interference pattern in a visible way. Therefore, when we immobilize the drum, the subtle vibration of the drum head after each strike, i.e. displacement of particles on drum head, can be reflected by the interference patter.

Therefore, I designed a Michelson interferometer, as shown in Figure 9 below.

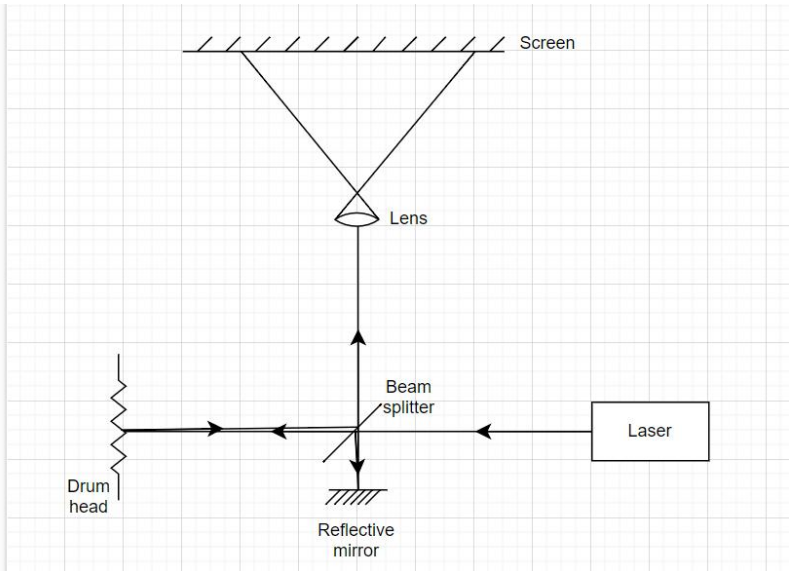


Figure 9. Design paper of the Michelson interferometer

After building it up, I initially tried replacing the drum head with another reflective mirror. By adjusting the angle of the mirror, I succeeded with superposing the two light spots, but the interference patter can hardly be seen, which is owing to the unstable installation of that mirror, as the interferometer is extremely sensitive to variations. Therefore, the first requirement of the interferometer is stability.

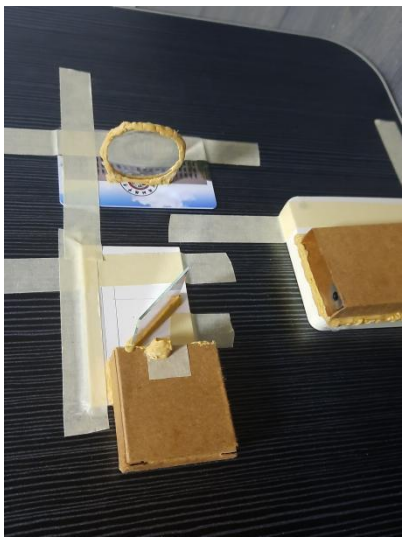


Figure 10. The installed Michelson interferometer (left), and the functioning system with drum head replaced by a reflective mirror.

After switching to the drum head again, another two problems appears. For one thing, the laser beam is too small to cover the entire drum head, so there must be another lens placing in between the beam splitter and drum head, allowing the beam to spread over the drum head. For another, the reflective index of drum head is significantly lower than the reflective mirror, causing a great intensity difference between the reference beam coming from the reflective mirror (left) and the beam coming from the drum head (right) as shown in Figure 11.

Figure 11. Intensity difference between the reference beam coming from the mirror (left) and the beam coming from the drum head (right)

Referencing to the three problems in the design and installation of the first Michelson interferometer, I redesigned another interferometer -- the Electronic Speckle Pattern Interferometry (ESPI).

Similar to Michelson's interferometer, ESPI is a non-contact, full-field optical measurement technique used to determine surface displacements, strains and deformations of an object. ESPI uses laser light to project a pattern of laser speckles onto a test object and then records the change in the speckle pattern due to any deformation in the surface. By analyzing the deformation in the speckle pattern, the ESPI technique is able to determine the surface displacement, strain, and other physical parameters of the object with high accuracy and precision.

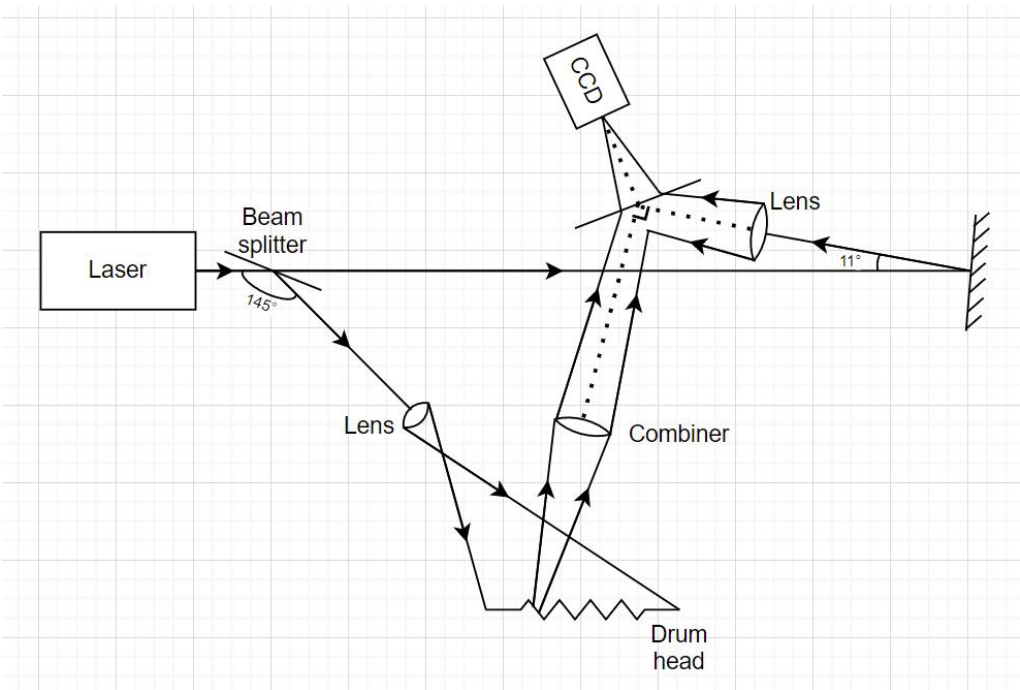


Figure 12. Design paper for the new ESPI system

See the design paper in Figure 12. For the stability problem, I rebuilt the entire system on a professional optical table, with most structures immobilized by screws instead of tapes. Also, by adding a lens between the beam splitter and the drum head, I enlarged the domain of the beam on the drum head. Moreover, I set a reflective mirror over

long distances from the light source, so that the reference beam must travel a longer distance, thus reducing its intensity.

4.2 Experiment procedures

Experiment equipment:

A diode laser, a beam splitter, two lens, two combiner, one reflective mirror, an acoustic drum, one charge-coupled device (CCD) camera.

Experiment procedure:

1. Assemble the optical system following the design paper in Figure 12.
2. Use the laser of shorter wavelength when the variation is relatively small, and longer wavelength if the variation is large. We use shorter wavelength here so that the relative small vibration in drum head can be seen.
3. Calibrate each lens and light path. Make sure that the laser collected from the combiner and the laser reflected by the mirror ultimately forms a right angle at the observing point.
4. Direct the CCD at right angle to the collecting mirror, and subtly fine tune the its angle so that the two light spots collected from the drum head and the reflective mirror eventually superpose on the CCD frame.

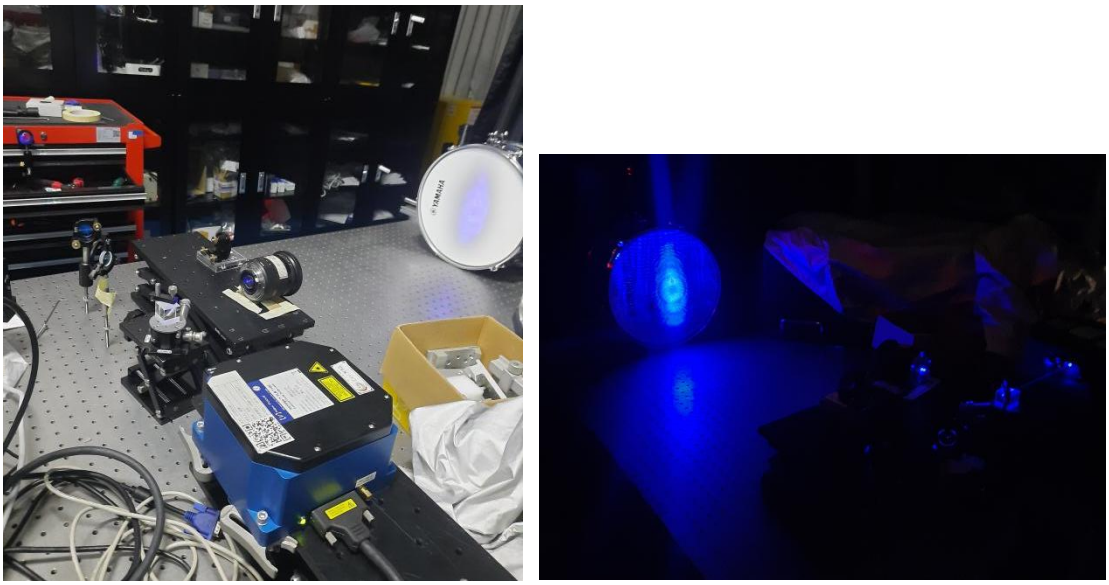


Figure 13. Installed ESPI system

4.3 Experiment results

Unfortunately, the intensity of the reference beam is still too large compared with the beam collected from the drum head, so the interference pattern cannot be clearly seen in the CCD as shown in Figure 14.

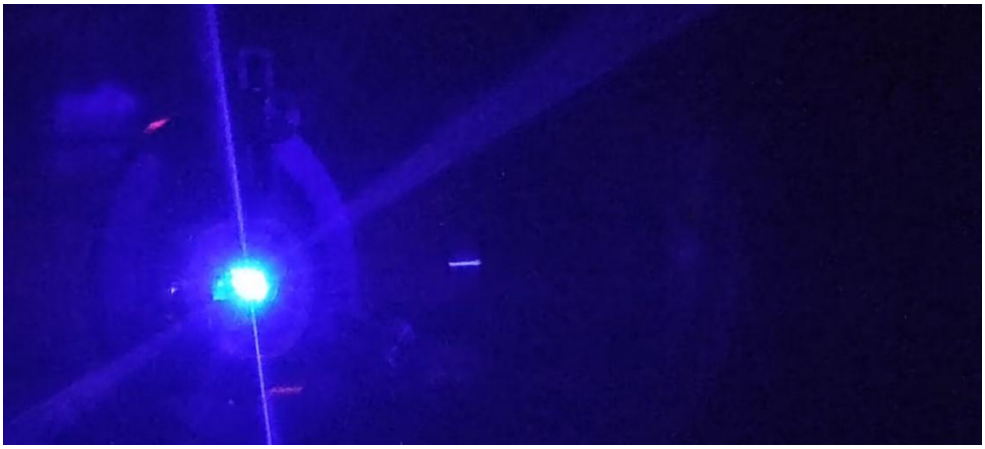


Figure 14. CCD image from ESPI experiment

A polarizing film that can effectively control the distribution of light intensity from the laser without energy loss can resolve the intensity difference problem, but the one

I got during the experiment was not properly functioning. Therefore, the result, unfortunately, cannot be derived from the experiment.

5. Conclusion and evaluation

The process of formula derivation, FEA simulation, and experimental investigation has led us to arrive at several interesting conclusions.

5.1 Progress conclusion

First, the correlation between surface tension and vibration mode is proven because surface tension will exclusively influence the angular velocity of the wave propagation in the membrane, which is one factor in the membrane particle displacement formula [2.1].

From simulation in *COMSOL Multiphysics*, we have arrived at a result that in terms of a single piece of batter head, the changing surface tension can vary the number of nodal circle n on the membrane -- the tighter the drum head tension, the smaller the number of nodal circles.

Second, theoretically, the three variables surface tension, vibration eigenfrequency, and relative frequency (related to vibration mode) are interconnected as Equation [3.3] shows. To be more specific, for one thing, under constant tension, the eigenfrequency of vibration will be proportional to the relative frequency of the corresponding

vibration mode; for another, under the same vibration mode, the eigenfrequency will be proportional to the square root of tension.

From the simulation results in *COMSOL Multiphysics*, the first point is verified as a positive correlation does appear between the relative frequency and eigenfrequency for every individual vibration. However, the second point is only partially proved, as the relationship is obvious between the 11500 and 23000 N m^{-1} group, but not between the 23000 and 46000 N m^{-1} group. Also, except for the eigenfrequency, the vibration frequency after excitation of drum head is also simulated in *COMSOL Multiphysics*, and we have proven the positive correlation between the surface tension and the period of vibration, i.e. a negative correlation between surface tension and vibration frequency.

In terms of the experiment, a Michelson interferometer and an ESPI system have been designed and built, although the problem of the huge intensity difference between the reference beam and the collected beam from drum head diffuse reflection has led to the failure of witnessing an interference pattern.

5.2 Evaluation and reflection

Part of the uncertainty comes from the limitation of the simulation software. The vibration of the membrane is presented frame by frame, and the minimum frame number the software can reach is a time interval of 0.0205 seconds, but this is still not enough to exhibit the movement of the membrane to a satisfyingly clean extent. For

one thing, the vibration mode is altering every one to two frames, so it can be speculated that some performed vibration modes are missing from the frames. For another, since the magnitude of tension is proportional to the magnitude of frequency, the vibration period of the three plates exhibited in Figure 4 may be different. Therefore, possibility exists for mismatching the three plates' vibrational modes from different period. (For reference, the image of the first seven frames are posted in Appendix 3)

For the ESPI experiment, an interference pattern would be possibly witnessed if the intensity difference between the reference beam and the collected beam from drum head's diffuse reflection is smaller. This can be achieved by either using a polarizing film to weaken the reference beam, or change to a combining lens that has a larger diameter, thus collecting more light from the drum head.

5.3 Future research possibilities -- Coupling vibration mode

Musical drums such as snare drum consist of two drum heads and a cylindrical air cavity in between. When striking the batter head of the drum, both heads of the drum will vibrate. This phenomenon is known as the coupling vibration of drums.

The investigation in this essay has already proven that after excitation on the batter head, it will oscillsate in a certain mode, put the air cylinder into resonance, then drive the opposite head to oscillate in a certain mode as well.

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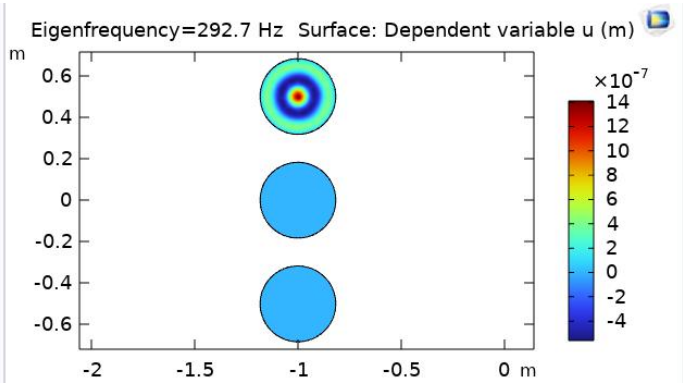
Appendix 1. Matlab codes for drawing Bessel function

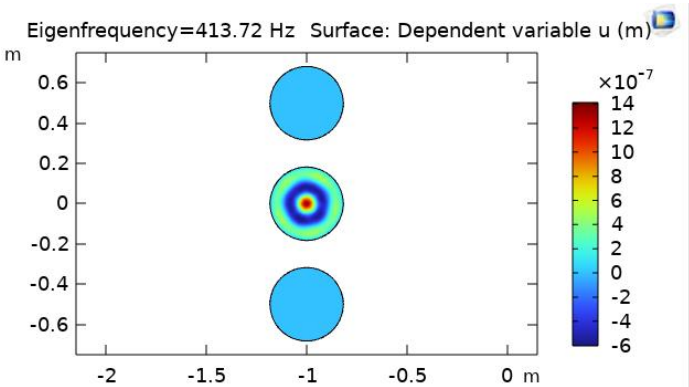
```
clear ,clc;

format long

x=(0:0.01:20)';

y_0=besselj(0,x);

y_1=besselj(1,x);

y_2=besselj(2,x);

plot(x,y_0,x,y_1,x,y_2);grid on;

axis([0,20,-1,1]);

title('Bessel functions of the first kind');

xlabel('x');

ylabel('Jm(x)');
```

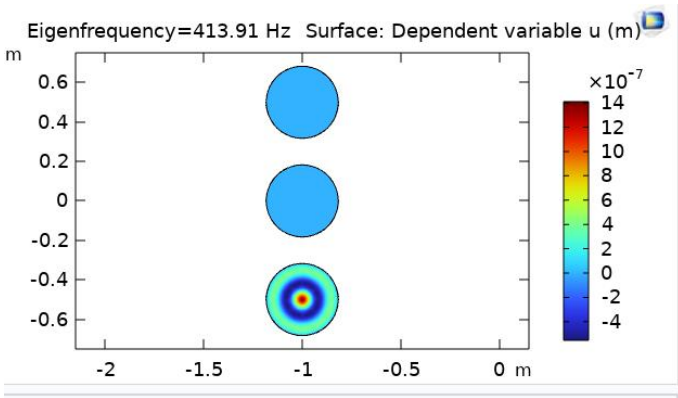
Appendix 2. Vibration modes with eigenfrequencies in three tension-varying plates

In the essay, an example of a combined vibration mode (2,3) is exhibited, so another two typical examples of vibration modes with exclusive circular and diameter node will be given here, along with their vibration eigenfrequency under three tensions:

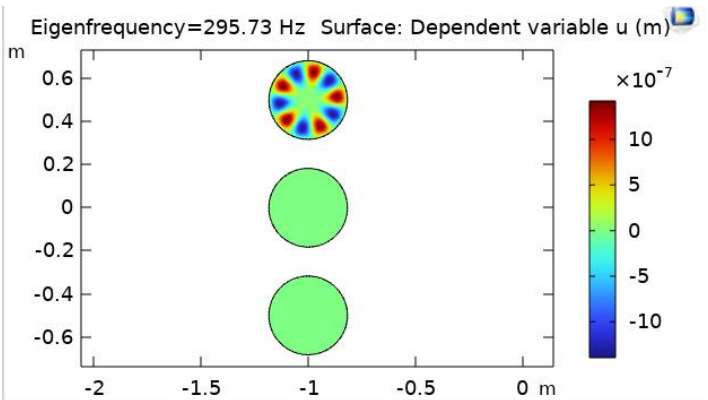
The (0,3) mode:

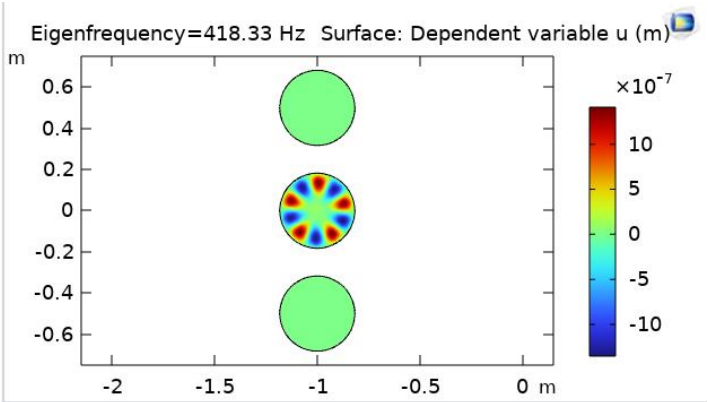

Eigenfrequency=292.7 Hz Surface: Dependent variable u (m)
m
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-2
-1.5
-1
-0.5
0 m
×10^-7
14
12
10
8
6
4
2
0
-2
-4

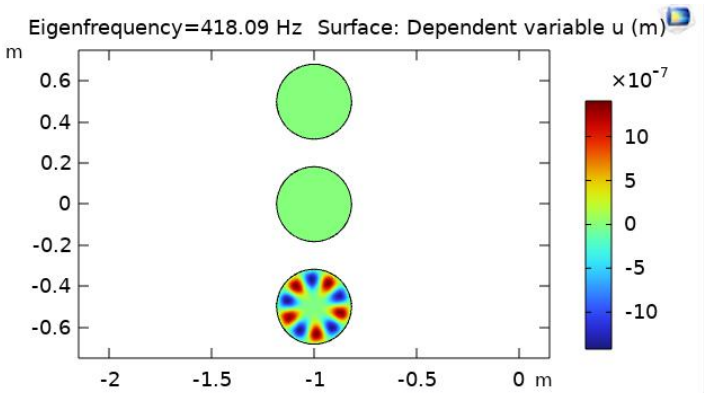

Eigenfrequency=413.72 Hz Surface: Dependent variable u (m)
m
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-2
-1.5
-1
-0.5
0 m
×10^-7
14
12
10
8
6
4
2
0
-2
-4
-6


Eigenfrequency=413.91 Hz Surface: Dependent variable u (m)
m
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-2
-1.5
-1
-0.5
0 m
×10^-7
14
12
10
8
6
4
2
0
-2
-4

The (5,1) mode:

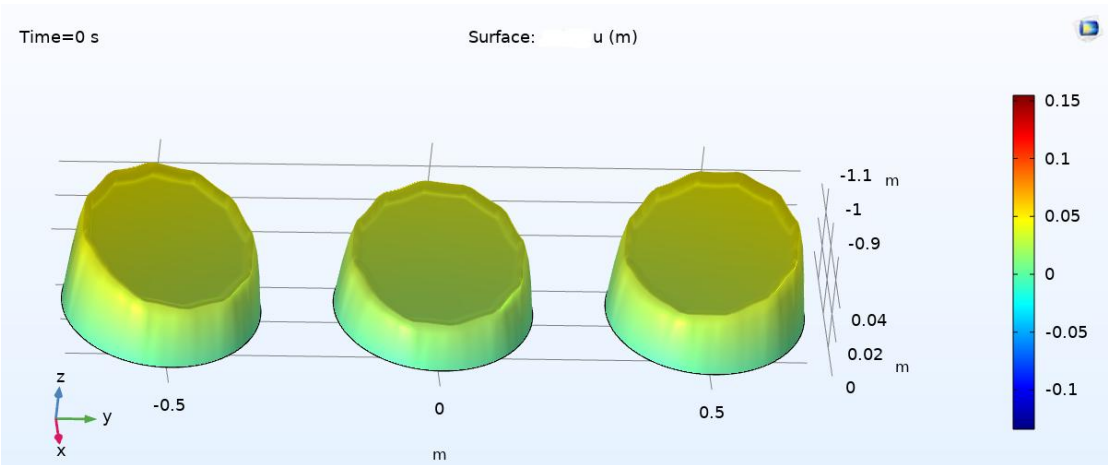

Eigenfrequency=295.73 Hz Surface: Dependent variable u (m)
m
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-2
-1.5
-1
-0.5
0 m
×10^-7
10
5
0
-5
-10



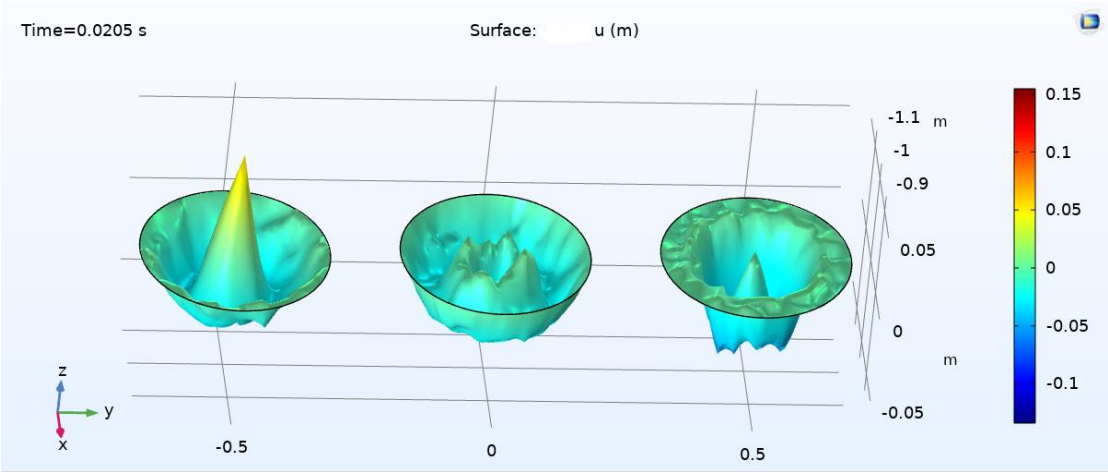


Appendix 3. The first seven frames of *COMSOL* simulation

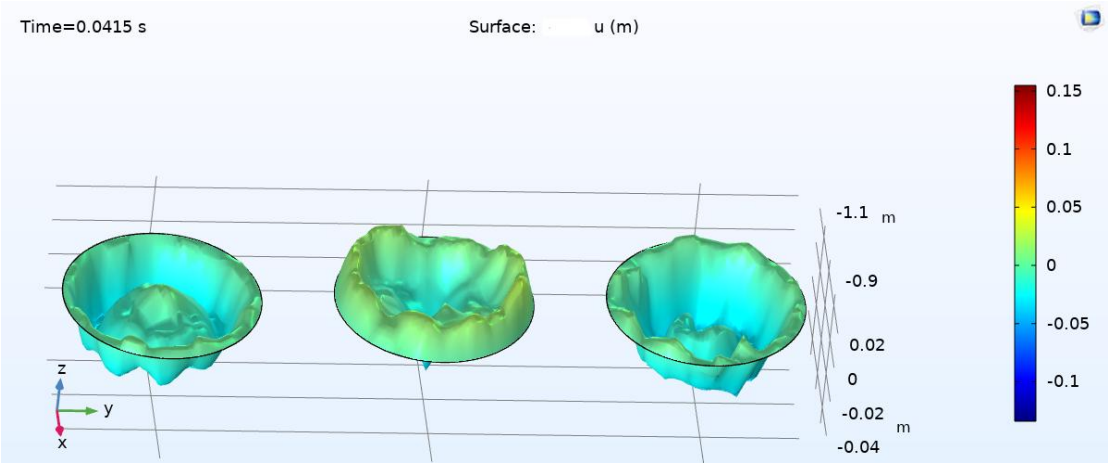
1st frame:



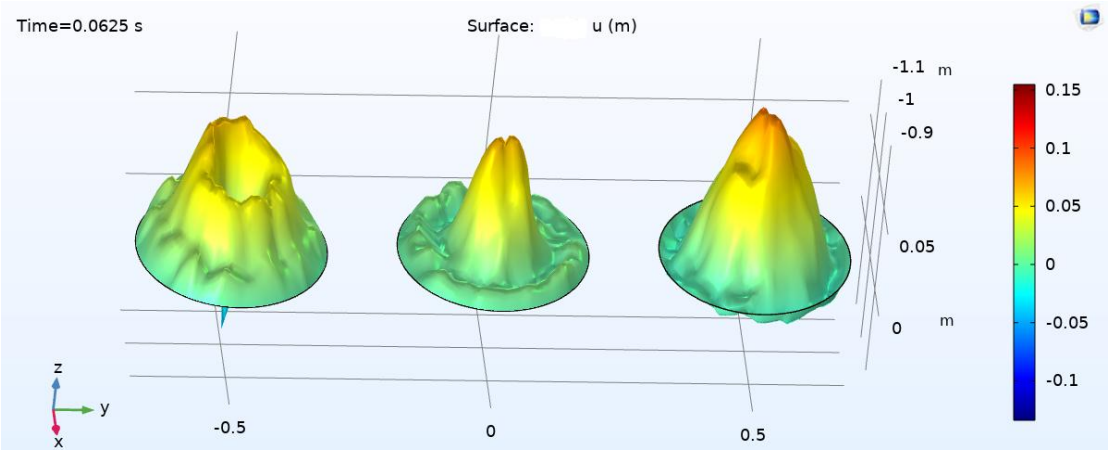
2nd frame:



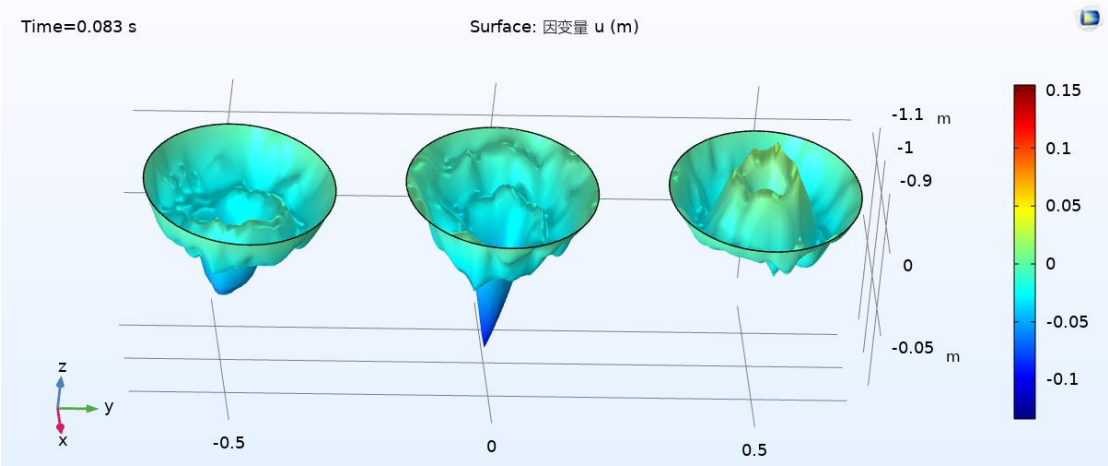
3rd frame:



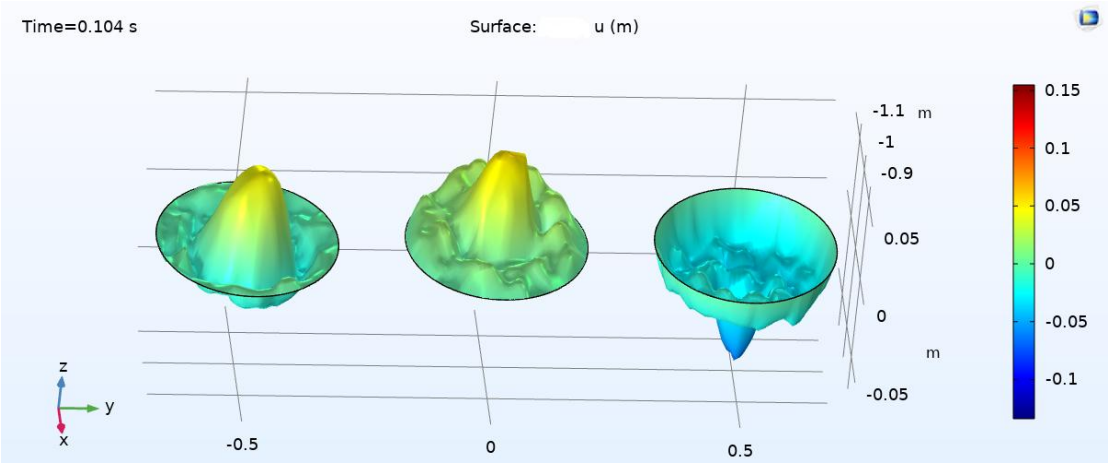
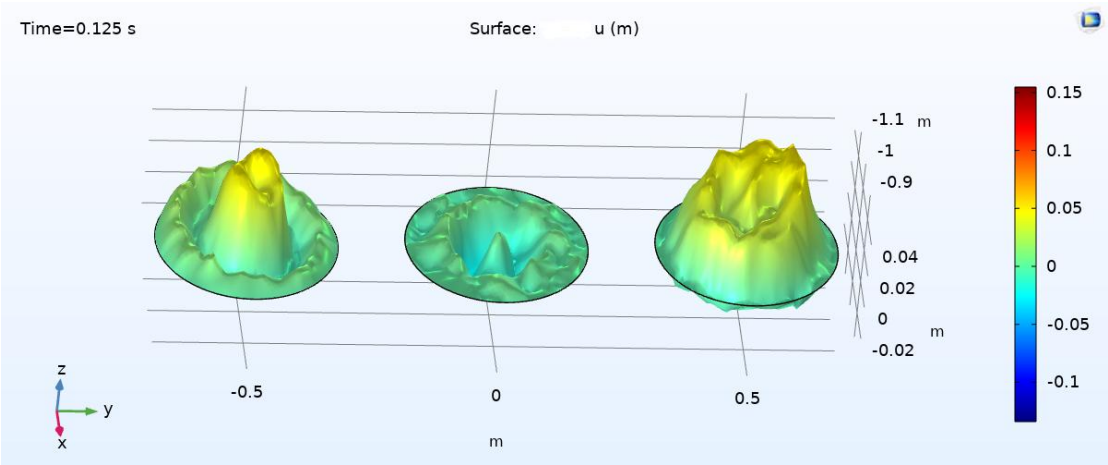
4th frame:



5th frame:



6th frame:



7th frame: